Cover letter

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Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “Two conjectures of Wormald on Hamilton cycles in tournaments” by Deep Bhattacharjee.

The manuscript proves Wormald’s Conjectures 2 and 6: a determinant formula for every regular tournament gives the rational-square property, and an exact cyclic formula gives the limit cosh(1)/cos(1). This revision details the analytic Hessian identification, proves quantitative positive-definiteness and supplies a direct spectral-radius argument.

The exact tournament invariant, cyclic formulas and algebraic proof fit the journal’s combinatorial and algebraic scope. The newer global Hamilton-cycle asymptotic of Divoux and Wang is explicitly credited; the manuscript does not claim that broader theorem as its contribution.

The prior public preprint is https://hexagonmath.org/2610.00460v1. The research manuscript/source archive is https://doi.org/10.5281/zenodo.23567153. Editable sources and verification records are supplied with the manuscript.

The unchanged Lean proof tree has a complete successful build and 23 axiom reports using only propext, Classical.choice and Quot.sound. It formalizes the determinant algebra and cyclic limit after identifying H with det(E^T F E). The analytic identification, asymptotic expansion, spectral bounds and new quantitative corollary have written proofs and are not claimed as Lean-formalized. Fresh exact checks verify 41 full Hessian matrices (3136 entries) and their bounds, and 142 manuscript values; numerical checks complement the universal proofs.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

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